

Abstract

This paper builds on linguistic case theory and cognitive science and provides a unified algebraic framework for cognitive systems modeling centered on the *case diagram*. Formally, case systems are modeled as categories whose objects are case roles and whose morphisms are grammatical relations; alignment typologies (nominative-accusative, ergative-absolutive, active-stative, tripartite, and fluid-S) are captured as structure-preserving functors between these categories. Within this framework, sentence structure becomes a compact-closed string diagram—computable via DisCoCat and, at the discourse level, DisCoCirc [coecke2010mathematical; defelice2022discocirc]—that functions not merely as descriptive notation but as an executable witness of grammaticality grounded in situated reasoning. Historically, the analysis traces from Pāṇinian *kāraka* theory [kak1987paninian] through Jakobson's feature matrices [jakobson1958morphologic] and Fillmore's deep cases [fillmore1968case] to Dowty's graded proto-roles [dowty1991thematic], showing how morphosyntactic inflection